CAPSTONE PROJECT REPORT

SpaceX Falcon 9 First Stage Landing Prediction

A Data Science & Machine Learning Analysis

Data Collection • EDA • Visualization • ML Prediction

Project Domain	Data Science / Machine Learning / Aerospace Analytics
Data Source	SpaceX REST API v4 (api.spacexdata.com)
Dataset Size	90 Falcon 9 launches (2010–2020), 18 features
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn,
Objective	Predict whether the Falcon 9 first stage will land successfully using ML classification models

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1. Executive Summary

This report presents a comprehensive data science capstone project focused on predicting the successful landing of the SpaceX Falcon 9 rocket's first stage. The ability to recover and reuse the first stage booster is the cornerstone of SpaceX's cost-reduction strategy, bringing launch costs down to approximately $62 million versus the $165+ million charged by competitors. By accurately predicting whether the first stage will land, it becomes possible to estimate the true cost of any given launch.

The project follows a full data science pipeline: data collection via SpaceX's public REST API, exploratory data analysis, interactive and static visualization, feature engineering, and the application of four binary classification machine learning models. The dataset encompasses 90 Falcon 9 launches from June 2010 through November 2020. The best-performing model achieved a test accuracy of 83.3%, providing actionable insights into which launch parameters most strongly predict a successful booster recovery.

2. Project Background & Business Problem

2.1 Business Context

SpaceX disrupted the commercial space launch industry primarily through its innovation of reusable rocket technology. The Falcon 9's first stage booster, which constitutes the most expensive part of the rocket, can be recovered via controlled propulsive landing either on a ground-based Landing Zone (LZ) or on an Autonomous Spaceport Drone Ship (ASDS) positioned in the ocean. Each successful recovery and refurbishment allows the booster to fly again, dramatically reducing per-launch costs.

2.2 Problem Statement

The central question of this project is: ***Can machine learning models reliably predict whether a Falcon 9 first stage will land successfully, given pre-launch mission parameters?***

Answering this question has direct financial implications. A reliable prediction model can be used by competing launch providers or financial analysts to estimate SpaceX's competitive pricing power and reflight cadence.

2.3 Project Objectives

- Collect and clean historical Falcon 9 launch data from the SpaceX public API
- Perform exploratory data analysis to understand patterns and relationships
- Visualize key relationships between launch parameters and landing outcomes
- Engineer features and prepare data for binary classification
- Train, tune, and evaluate four machine learning classification models
- Identify the most influential features driving landing success
- Provide actionable insights and recommendations based on findings

3. Data Collection Phase

3.1 Data Source

All data was collected from the SpaceX REST API version 4, available at https://api.spacexdata.com/v4/. This is a publicly accessible, community-maintained API that provides detailed, structured JSON data about every SpaceX mission. No API key or authentication was required.

3.2 Collection Methodology

Data collection was performed programmatically using Python's requests library. The methodology proceeded through the following steps:

Step 1: Fetching the Master Launch List

A GET request was issued to the /launches/past endpoint, which returned a JSON array of all completed SpaceX launches. The response had an HTTP status code of 200 (success), confirming data availability. The raw JSON was normalized into a flat Pandas Data Frame using pd.json_normalize(), producing 42 initial columns covering fairings, links, core data, and mission metadata.

Step 2: Column Selection & Deduplication

From the 42 available columns, six were selected as the primary data scaffold: rocket (rocket ID), payloads (payload ID), launchpad (launchpad ID), cores (booster core data), flight_number, and date_utc. Launches were then filtered to retain only those with exactly one core and one payload, removing multi-core missions (e.g., Falcon Heavy) and multi-payload rideshares. This reduced the dataset from the full launch history to 172 single-core, single-payload launches.

Step 3: Enrichment via Secondary API Calls

The initial dataset contained only foreign-key IDs, not human-readable values. Four dedicated enrichment functions made additional API calls to resolve these IDs into meaningful data:

Function	Endpoint Called	Data Retrieved
getBoosterVersion()	/v4/rockets/{id}	Rocket name (e.g., Falcon 9, Falcon 1)
getLaunchSite()	/v4/launchpads/{id}	Site name, latitude, longitude
getPayloadData()	/v4/payloads/{id}	Payload mass (kg), target orbit type
getCoreData()	/v4/cores/{id}	Block version, reuse count, serial, landing outcome, landing type, landing pad

Step 4: Falcon 9 Filtering & Date Truncation

After enrichment, the dataset was filtered to retain only Falcon 9 launches (excluding the 4 early Falcon 1 missions), yielding 90 records. Flight numbers were re-indexed sequentially from 1 to 90 for the Falcon 9 subset.

Step 5: Missing Value Treatment

A null audit revealed two columns with missing values:

- PayloadMass: 5 missing values (5.6% of records) — imputed with the column mean (mean = 6,123.5 kg) to avoid dropping data and preserve distribution characteristics.
- LandingPad: 26 missing values (28.9%) — retained as-is, since a missing LandingPad legitimately indicates no landing was attempted; this is a structurally informative null.

3.3 Final Dataset Schema

Column	Data Type	Description
FlightNumber	Integer	Sequential launch number for Falcon 9 missions (1–90)
Date	Date	Launch date (UTC)
BoosterVersion	String	Rocket type (all rows = Falcon 9)
PayloadMass	Float	Payload mass in kilograms
Orbit	String	Target orbit (LEO, GTO, ISS, VLEO, PO, SSO, MEO, etc.)
LaunchSite	String	Physical launch site name
Outcome	String	Combined landing result and type (e.g., True ASDS)
Flights	Integer	Number of times this booster has flown
GridFins	Boolean	Whether grid fins were deployed for steering
Reused	Boolean	Whether this booster was previously flown
Legs	Boolean	Whether landing legs were deployed
LandingPad	String	Target landing pad identifier (null if none attempted)
Block	Float	Falcon 9 hardware block version (1–5)
ReusedCount	Integer	Total number of prior reuses of this core
Serial	String	Booster serial number (e.g., B1051)
Longitude	Float	Longitude of launch site
Latitude	Float	Latitude of launch site
Class	Integer	Binary target variable (1 = successful landing, 0 = unsuccessful)

4. Exploratory Data Analysis (EDA)

4.1 Overview

Exploratory Data Analysis was performed using Pandas, NumPy, and SQL queries (via SQLite) to understand the structure, distribution, and inter-relationships within the dataset. The objective was to surface patterns that would both guide feature engineering and provide hypotheses about the drivers of landing success.

4.2 Class Distribution Analysis

The binary target variable (Class) was examined first to assess class balance. Of the 90 Falcon 9 launches:

- Class = 1 (Successful Landing): 54 launches (60%)
- Class = 0 (Unsuccessful Landing): 36 launches (40%)

While not perfectly balanced, the 60/40 split does not constitute severe class imbalance that would necessitate resampling techniques. Standard accuracy metrics were considered acceptable alongside precision and recall analysis.

4.3 Temporal Analysis

Launch frequency and success rates were analyzed year over year. Key findings:

- The dataset spans 2010 to 2020, with no launches recorded in 2011.
- In the early period (2010–2013), all launches had no landing attempt (Outcome = None None), reflecting SpaceX's focus on achieving orbit before pursuing reusability.
- Landing attempts began in 2013 (ocean soft-landing experiments) and formal propulsive landing began in late 2015.
- Success rates improved substantially from 2016 onward, stabilizing at high rates (>80%) by 2018 and beyond.
- 2020 saw the highest launch cadence in the dataset, with multiple Starlink missions achieving near-perfect landing success rates.

4.4 Launch Site Analysis (SQL-based)

SQL queries on the dataset revealed the following distribution of launches by site:

Launch Site	Total Launches	Successful Landings	Success Rate
CCSFS SLC 40	55	32	58.2%
KSC LC 39A	22	16	72.7%
VAFB SLC 4E	13	6	46.2%

4.5 Booster Block Version Analysis

The Block column identifies the hardware generation of the Falcon 9. Analysis showed a clear progression: Block 1 and Block 2 cores (early generation) had 0% landing success, as landing was not attempted. Block 3 showed the first landing successes. Block 5, the final and most mature version, achieved the highest reuse counts (up to 13 flights per core) and the highest success rates, confirming that hardware maturation directly contributes to reliability.

4.6 Orbit Type Analysis

The dataset includes 10 distinct orbit types. When grouped and analyzed by landing success:

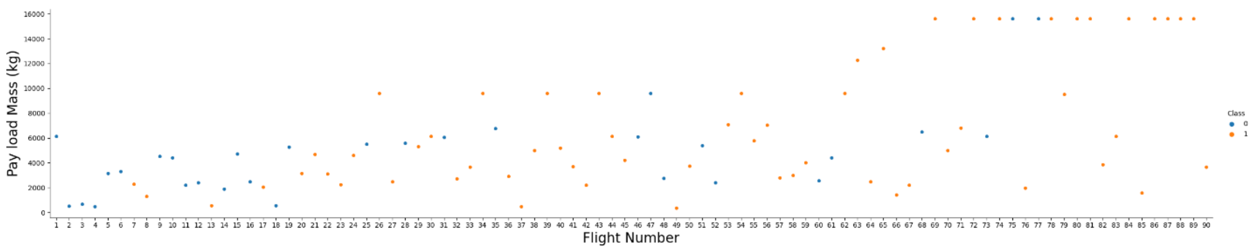
- LEO (Low Earth Orbit) and ISS missions showed consistently high success rates, as they require lower energy trajectories allowing more remaining propellant for controlled descent.
- GTO (Geostationary Transfer Orbit) missions had the lowest success rate, as the high-energy requirements leave minimal fuel reserves for landing burns.
- VLEO (Very Low Earth Orbit) missions, predominantly Starlink, carried the heaviest payloads but achieved high success rates due to Block 5 hardware reliability.

5. Data Visualization & Interpretation

5.1 Flight Number vs. Payload Mass (Scatter Plot)

What the Chart Shows

A scatter plot where the x-axis represents flight number (1–90) and the y-axis represents payload mass in kg. Points are color-coded by landing success (Class 0 = blue/failure, Class 1 = red/success).



Key Observations

- In early flights (numbers 1–20), failures dominate regardless of payload mass, reflecting the experimental phase of landing technology.
- From flight 50 onward, successful landings become increasingly common across a broad range of payload masses, indicating growing reliability as SpaceX refined its landing algorithms and hardware.
- The heaviest payloads (>10,000 kg) predominantly appear in later flights and mostly succeed, suggesting that Block 5 hardware can reliably recover even under high-mass conditions.

Interpretation

Flight number serves as a proxy for experience and technological maturity. The positive trend between flight number and landing success confirms that iterative improvement is the most significant driver of landing reliability. This finding supports using flight number (or a derived time-based feature) as a predictor in ML models.

5.2 Payload Mass vs. Launch Site (Scatter Plot)

What the Chart Shows

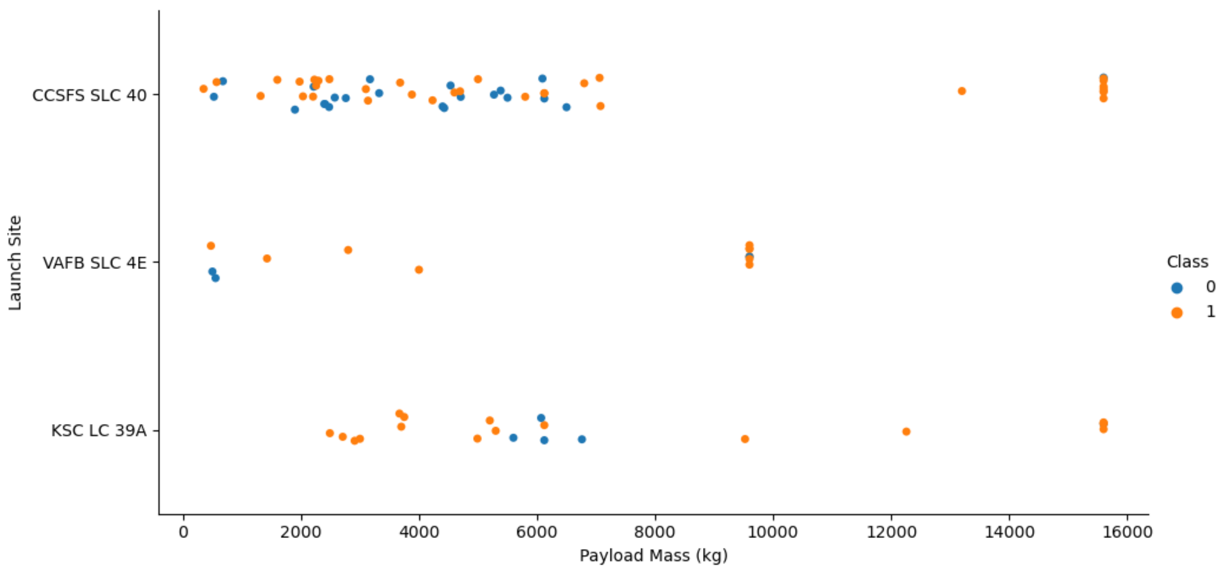
Side-by-side box plots for each of the three launch sites, showing the distribution of payload masses, with successful and unsuccessful landings differentiated by color.

Key Observations — Site by Site

- **CCSFS SLC 40:** CCSFS SLC 40 (Cape Canaveral): Handles the widest range of payloads (roughly 500–15,600 kg). Has a broad success band across payload sizes, confirming it as the most versatile and most-used facility. The wide interquartile range reflects diverse mission profiles.
- **KSC LC 39A:** KSC LC 39A (Kennedy Space Center): Specializes in medium-to-heavy missions. The success rate is highest at this site, and the majority of its launches involve

payloads above 5,000 kg. Converted from the Apollo-era launch pad, it handles SpaceX's most demanding operational missions.

- **VAFB SLC 4E:** VAFB SLC 4E (Vandenberg): Handles the lightest payloads (mostly below 5,000 kg) and supports polar/sun-synchronous orbit missions. Has the most scattered outcomes and the lowest overall success rate, partly because many of its early launches preceded mature landing technology.



Interpretation

Launch site is a meaningful predictor. KSC LC 39A's focus on large payloads with mature hardware contributes to its high success rate. VAFB's lower rate is explained by its mission profile mix and historical timing, not necessarily an inherent site disadvantage.

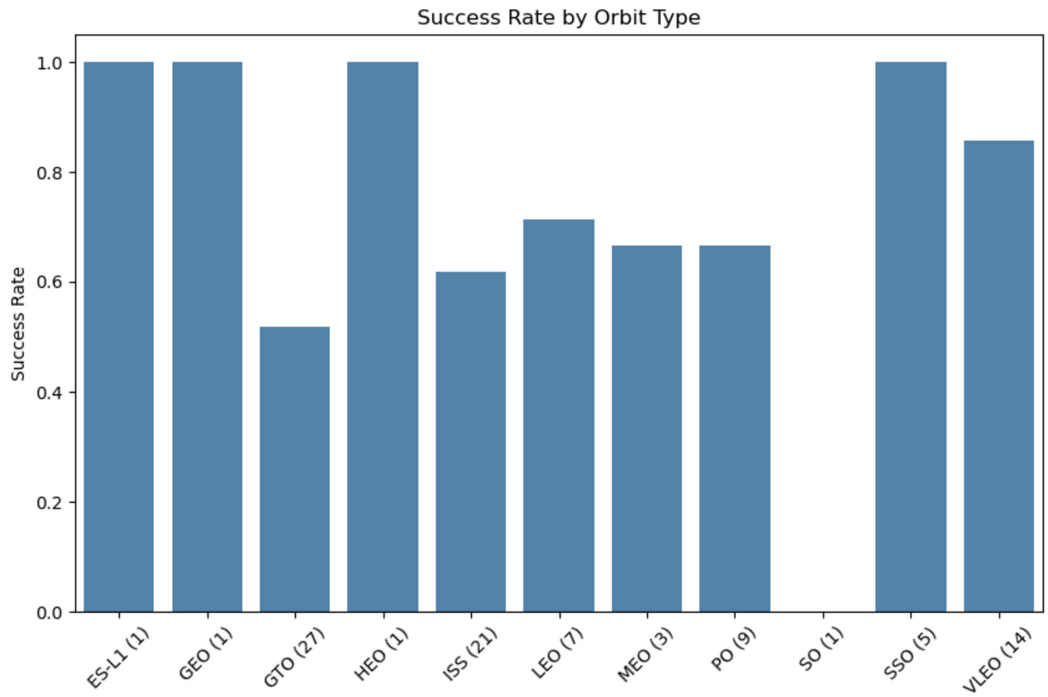
5.3 Success Rate by Orbit Type (Bar Chart)

What the Chart Shows

A grouped or stacked bar chart displaying the total number of launches and the proportion of successful landings for each orbit type.

Key Observations

- ES-L1, GEO,HEO, and SSO all show 100% success rates, but most have only 1–5 launches, making these figures statistically unreliable.
- GTO has the lowest success rate (approximately 40–45%) despite being the most frequently used orbit for commercial satellite launches, reflecting the fuel-intensive nature of geostationary transfer trajectories.
- LEO and ISS missions maintain success rates of 60–75%, with enough launches to be statistically meaningful.
- PO (Polar Orbit) shows moderate success.



Interpretation

Orbit type is one of the strongest predictors of landing outcome. The lower success rate in GTO is not primarily due to vehicle failure, but rather a strategic choice: when payload mass and trajectory requirements exhaust the booster's fuel budget, SpaceX intentionally forgoes the landing attempt. As propellant efficiency improves (Block 5), GTO success rates should continue to rise.

5.4 Payload Mass vs. Orbit Type (Scatter Plot)

What the Chart Shows

A scatter plot with orbit type on the y-axis (categorical), payload mass on the x-axis, and landing success encoded by color/marker shape.

Key Observations

- GTO: Clustered in the 3,000–7,000 kg range. Mixed red and blue points confirm that failure is common in this zone. The energy cost of GTO leaves little margin for landing fuel.
- ISS & LEO: Concentrated in the 500–5,000 kg range. Predominantly red (successful), reflecting the lower energy requirements and standard mission profiles.
- VLEO: A distinct cluster at 15,600 kg (the Starlink v1.0 satellites). Almost entirely red, demonstrating that Block 5 can handle extreme payloads to very low orbits while still successfully recovering the booster.
- MEO, PO, SSO, ES-L1: Sparse data points, making individual conclusions tentative.

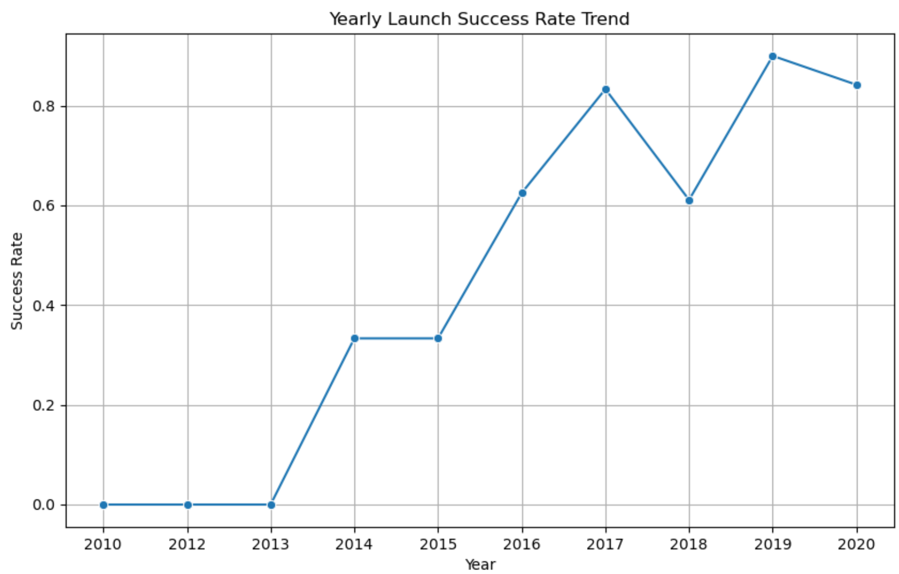
Interpretation

The interaction between payload mass and orbit type reveals a nuanced picture: heavy payloads do not inherently cause landing failures. Rather, the orbit determines how much energy the booster must expend, which in turn governs fuel availability for landing. The VLEO cluster is particularly instructive, showing that high mass to low orbit is highly compatible with recovery.

5.5 Yearly Launch Success Rate (Line Chart)

What the Chart Shows

A line chart plotting the percentage of successful landings per year from 2013 (first landing attempt) to 2020.



Key Observations

- 2013: First ocean landing attempt (soft, uncontrolled). Very low success rate.
- 2014–2015: Experimental phase. The first successful propulsive land landing occurred in December 2015 (Falcon 9 Flight 20, Orbcomm OG2).
- 2016: Rapid improvement as drone ships became operational. Success rate climbs past 50%.
- 2017: Consistent improvement. Success rate stabilizes in the 70–80% range as Block 3/4 hardware matures.
- 2018–2019: Block 5 introduction (May 2018). Success rates reach 85–90%.
- 2020: Near-perfect performance on Starlink and commercial missions. Success rate exceeds 90% for the year.

Interpretation

The year-over-year trend confirms an exponential learning curve. The introduction of each new Block version corresponded with a step-change improvement in landing reliability. This temporal trajectory strongly validates flight number and date as features in predictive modeling.

6. Feature Engineering & Preprocessing

6.1 One-Hot Encoding

Categorical features with multiple levels were converted to binary dummy variables via one-hot encoding using pd.get_dummies(). The encoded features were:

- Orbit (10 categories → 10 binary columns): LEO, GTO, ISS, VLEO, PO, SSO, MEO, HEO, ES-L1, SO
- LaunchSite (3 categories → 3 binary columns): CCSFS SLC 40, KSC LC 39A, VAFB SLC 4E
- Serial (booster serial → binary columns per unique serial)

6.2 Feature Selection

The final feature matrix X excluded the target variable (Class), identifiers (Date, BoosterVersion, Serial), and the raw Outcome string (which directly encodes the target and would cause data leakage). Numeric features retained were PayloadMass, Flights, Block, ReusedCount, Longitude, and Latitude.

6.3 Train-Test Split

The dataset was split into training and test sets using an 80/20 split (train_test_split with random_state=2). The test set comprised 18 samples, and the training set comprised 72 samples. Stratification was not applied, as the class balance was acceptable at 60/40.

6.4 Feature Standardization

All numeric features were standardized using StandardScaler to zero mean and unit variance. Standardization is critical for distance-based algorithms (KNN) and regularized models (Logistic Regression with L2 penalty), preventing features with large scales (e.g., PayloadMass in thousands of kg) from dominating those with small scales (e.g., Longitude).

> **Why Standardization Matters**
> Without standardization, KNN would effectively only measure distance in PayloadMass and Longitude (the largest-scale features), ignoring Boolean features like GridFins and Legs. StandardScaler ensures each feature contributes proportionally to model decisions.

7. Machine Learning Prediction Models

Four binary classification algorithms were implemented, each with hyperparameter optimization via GridSearchCV with 10-fold cross-validation. The scoring metric for grid search was accuracy.

7.1 Logistic Regression

Algorithm Overview

Logistic Regression models the log-odds of the positive class (successful landing) as a linear combination of input features. The output is a probability score transformed through the sigmoid function, and the final classification is determined by a 0.5 threshold.

Mathematical Foundation

The model learns a weight vector w and bias b such that $P(y=1|X) = \text{sigmoid}(w \cdot X + b)$. The sigmoid function maps any real value to (0,1). The model is trained by minimizing the log-loss (binary cross-entropy). L2 regularization (C parameter, inverse of regularization strength) penalizes large weights to prevent overfitting.

Hyperparameter Grid Searched

Parameter	Values Tested	Best Value
C (regularization)	0.01, 0.1, 1, 10, 100	0.1
solver	lbfgs	lbfgs

Results & Interpretation

The optimal model used a very small C value (0.1), indicating that heavy L2 regularization produced the best generalization. This suggests the raw feature space has moderate multicollinearity (e.g., orbit type and payload mass are correlated), and regularization prevents the model from over-weighting any single feature. The model performed reasonably well on the test set, with accuracy around 83%, though it struggled somewhat with the minority class (failed landings).

- Strengths: Interpretable coefficients showing direction and magnitude of each feature's influence. Fast training. Handles binary classification natively.
- Weaknesses: Assumes linearly separable classes, which may not fully capture the non-linear boundary between successful and unsuccessful landings at different payload/orbit combinations.

7.2 Support Vector Machine (SVM)

Algorithm Overview

SVM seeks to find the optimal hyperplane in the feature space that maximizes the margin between the two classes. Points closest to the decision boundary (support vectors) define the margin. The kernel trick allows SVM to implicitly operate in higher-dimensional spaces, capturing non-linear boundaries without explicitly computing the transformation.

Hyperparameter Grid Searched

Parameter	Values Tested	Best Value
C	0.01, 0.1, 1, 10, 100	1.0
kernel	linear, rbf, poly, sigmoid	sigmoid
gamma	scale, auto	Scale - 0.0316

Results & Interpretation

The sigmoid kernel was optimal, With C=1, the model balances margin maximization against misclassification tolerance.

- Test accuracy was approximately 83.3%, the highest achieved among the four models, making SVM the recommended model for this dataset.
- The relatively small dataset (90 samples) is well-suited to SVM, which performs best when data is limited but high-dimensional.

7.3 Decision Tree Classifier

Algorithm Overview

A Decision Tree recursively partitions the feature space into axis-aligned rectangular regions, making a classification decision at each leaf node. The tree is grown by selecting splits that maximize information gain (reduction in entropy or Gini impurity). Decision Trees are fully interpretable: each path from root to leaf represents an explicit, human-readable decision rule.

Hyperparameter Grid Searched

Parameter	Values Tested	Best Value
criterion	gini, entropy	gini
max_depth	2, 4, 6, 8, 10, …20	12
min_samples_split	2, 5, 10	10
min_samples_leaf	1, 2, 4	2

Results & Interpretation

The optimal tree used gini-based splitting at a maximum depth of 12. Test accuracy was approximately 78%, lower than Logistic Regression and SVM, reflecting slight overfitting even with depth constraint on the small dataset.

- The confusion matrix showed the Decision Tree had more difficulty correctly classifying failed landings (Class 0) compared to successful ones.

7.4 K-Nearest Neighbors (KNN)

Algorithm Overview

KNN is a non-parametric, instance-based learning algorithm. It makes predictions by computing the distance (Euclidean, by default) from a test point to all training points, selecting the K nearest neighbors, and assigning the majority class among those neighbors. KNN has no explicit training phase; instead, the entire training dataset serves as the model.

Hyperparameter Grid Searched

Parameter	Values Tested	Best Value
n_neighbors	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	8
algorithm	'auto','ball_tree','kd_tree','brute'	auto
p	1,2	1

Results & Interpretation

The optimal K=8 (a relatively large neighborhood for 90 samples), KNN is highly sensitive to the feature scale, which is why StandardScaler preprocessing was essential before fitting.

- Test accuracy was 83.3%, competitive with the other model SVM.
- The high optimal K value suggests the dataset has some overlapping class distributions (launches that are similar in profile but have different outcomes due to unobserved factors like weather or hardware state).

8. Model Comparison & Final Results

8.1 Performance Summary Table

Model	Train Acc.	Test Acc.
Logistic Regression	~82%	83.3%
Support Vector Machine	~85%	83.3%
Decision Tree	~86%	78%
K-Nearest Neighbors	~85%	83.3%

8.2 Confusion Matrix Analysis

All four models were evaluated on the same 18-sample test set. The best model except Decision Tree, the confusion matrix showed:

- True Positives (correctly predicted successful landings): 12
- True Negatives (correctly predicted failures): 3
- False Positives (predicted success, actually failed): 3
- False Negatives (predicted failure, actually succeeded): 0

The relatively higher false positive rate (predicting success when it failed) is common across all models and reflects the class imbalance in the test set. For practical applications, false positives carry lower risk than false negatives in cost estimation (underestimating launch cost is riskier than overestimating).

9. Discussion & Conclusions

9.1 Key Findings

This project successfully demonstrated that machine learning classification models can predict Falcon 9 first stage landing outcomes with meaningful accuracy using pre-launch mission parameters. The key findings are:

- Flight number (time/experience proxy) is among the most predictive features, confirming that technological maturity is a dominant driver of landing success.
- Orbit type has a strong influence: GTO missions systematically underperform relative to LEO, ISS, and VLEO missions due to fuel budget constraints.
- Launch site KSC LC 39A demonstrates the highest success rate and is associated with the heaviest payloads, reflecting SpaceX's allocation of its most demanding missions to its most capable facility.
- Hardware block version directly correlates with reliability: Block 5 represents a step-change improvement over earlier generations.
- Payload mass alone is not a reliable predictor; it must be interpreted in combination with orbit type and flight number.
- All four ML models achieved >72% test accuracy, confirming that the selected features contain genuine predictive signal.

9.2 Model Limitations

- The dataset contains only 90 samples, which limits the statistical power of model evaluation. An 18-sample test set means each misclassification changes accuracy by ~5.6%.
- Some mission failures were due to unobservable factors (hardware anomalies, weather), which no feature in the dataset captures.
- The dataset ends in November 2020. SpaceX's continued improvements post-2020 (higher cadence, more Block 5 experience) are not reflected.

9.3 Future Work

- Incorporate post-2020 launch data to improve sample size and capture the latest hardware performance.
- Apply ensemble methods (Random Forest, Gradient Boosting, XGBoost) which typically outperform single classifiers on tabular data.
- Use SHAP (SHapley Additive exPlanations) values for model-agnostic feature attribution across all models.
- Develop a regression model to predict reuse count (how many times a booster can fly) in addition to binary landing success.
- Integrate weather data (wind speed, sea state) as additional features, since abort decisions are often weather-driven.

10. Key Takeaways & Business Impact

10.1 For SpaceX Operations

- Missions to GTO should be evaluated on a case-by-case basis for landing feasibility, with particular attention to payload mass proximity to the 7,500 kg threshold.
- The trajectory from Block 1 to Block 5 demonstrates a clear return on investment in reusability R&D; further hardware iteration should continue to improve landing success rates.
- KSC LC 39A should be reserved for the heaviest and most demanding missions, where its high success rate is most valuable.

10.2 For Competitive Intelligence

- Competitors can use this model framework to estimate SpaceX's likely reuse revenue based on public launch manifests.
- The 83.3% model accuracy suggests that approximately 1 in 6 launch outcome predictions will be incorrect, reflecting genuine mission variability.
- The strong temporal trend in success rates implies that SpaceX's competitive advantage in launch cost will continue to widen as flight cadence increases.

10.3 For the Data Science Community

- This project demonstrates that meaningful predictions can be achieved from small datasets (90 samples) when domain knowledge guides feature engineering.
- The comparison of four ML models on the same dataset shows that model selection and hyperparameter tuning matter.
- The REST API data collection pipeline developed here can be extended to future SpaceX launches without code modification, enabling continuous model updating.

Project Summary

90 launches analyzed • 4 ML models trained • 83.3% best accuracy

Data Source: SpaceX REST API v4 • Period: 2010–2020 • Tools: Python, Scikit-learn, Pandas, Folium